

TEST PLAN

Login & roles

- ✓ Wrong password is rejected; shows “incorrect username or password.”
- ✓ After login, the dashboard matches the role (student/instructor/admin).

Student

- ✓ See catalog with code/title/credits/capacity/instructor.
- ✓ Register in a section with free seats → success and appears in timetable.
- ✓ Try to register the same section again → blocked with a clear message.
- ✓ Try to register in a full section → blocked with “Capacity Full.”
- ✓ Drop before deadline → success; appears removed.
- ✓ View grades for registered courses and sections.
- ✓ Download transcript (CSV) listing completed courses/grades.

Instructor

- ✓ See only their own sections.
- ✓ Enter scores (quiz/midterm/end-sem) → saved.
- ✓ Compute final grade using the rule in the report → shows final values.
- ✓ Try to edit a section they do not teach → blocked.

Admin

- ✓ Create a new student user (username+role in Auth DB; student profile in ERP DB).
- ✓ Create a course and a section; assign instructor.
- ✓ Toggle Maintenance ON → banner shows; student/instructor can view but cannot change methods.
- ✓ Toggle Maintenance OFF → normal behavior returns.

Password & Auth separation

- ✓ Password hash exists in Auth DB (not the real password).
- ✓ ERP DB does not contain passwords.
- ✓ Login uses Auth DB, then loads student/instructor profile from ERP DB via the shared user id.

Exports (pick at least one)

- ✓ Transcript export (Student) and grade CSV export (Instructor) works and opens.

Edge & negative tests

- ✓ Capacity cannot be negative or nonsensical.
- ✓ Register/drop after the stated deadline → blocked with message.
- ✓ Student cannot view or change another student’s data.
- ✓ Instructor cannot grade students from another instructor’s section.
- ✓ With Maintenance ON, student/instructor changes are blocked everywhere

Data integrity checks

- ✓ Duplicate enrollments (same student+same section) are prevented.
- ✓ Removing a section: either blocked when students are enrolled or clearly explained/documented.

Security basics

- ✓ Real passwords are never stored; only hashes.
- ✓ A Change Password dialog exists (bonus).
- ✗ After 5 wrong login attempts, show a warning or temporary lock (bonus).

UI/UX checks

- ✓ Buttons and labels are clear; errors are friendly.
- ✗ Tables for lists; sorting/filtering is a plus.
- ✓ Long actions show “please wait” (even a simple dialog). (Mostly all the actions are implemented immediately.)

Performance sanity

- ✓ Catalog list of ~100 courses opens quickly (a few seconds or less).
- ✓ App starts without long hangs or crashes.
- ✓ Maintenance & (optional) backup/restore
- ✓ Maintenance toggle flips the settings flag and the UI shows a banner immediately.
- ✓ All student/instructor writes are blocked while ON.
- ✗ (Bonus) Backup/restore: trigger backup, change a row, restore, and see it revert.

DATABASE SCREENSHOTS

```
MySQL 8.0 Command Line Cli x + v
+-----+
| Tables_in_auth_db |
+-----+
| users              |
+-----+
1 row in set (0.01 sec)

mysql> select * from users;
+-----+-----+-----+-----+-----+
| id | username | password | role | last_login |
+-----+-----+-----+-----+-----+
| 1 | admin | admin123 | ADMIN | NULL |
| 2 | chaitanya | yo | student | NULL |
| 3 | anubha | tough | instructor | NULL |
| 4 | advik | yo | student | NULL |
| 5 | david | qwerty | student | NULL |
| 6 | HAALAND | MCI | instructor | NULL |
| 7 | admin2 | $2a$10$MFwj7./o47IvangjJ3L.m0JY79YF1HuQs3g00q2gjbM0pEUWfTpj2 | admin | 2025-11-27 20:11:31 |
| 8 | dennispro | $2a$10$w8Ysff0Js/NuSSVCvvZ0jOI3Y9L6N0oC6xNNMX3I/MJV57Vu1EIQ | student | 2025-11-21 02:56:20 |
| 9 | profdavid | $2a$10$00VNQfCedgibCWmcRW2eFu6G00L30hxzW85CLZyaqR1.xnVjlcLt6 | instructor | 2025-11-27 20:12:00 |
| 10 | chaitanya | $2a$10$9rt59n3vJY.hdwFdvN/uzOU2UcVeTejw9ps0BdFWI2iRsuNIbS512 | student | NULL |
| 11 | advik | $2a$10$KLDWuqoqGcGcICmS1E1I9uGoGYgx/WMnAAgfgMo.PTLDcyEsubaWq | student | NULL |
| 12 | krishna | $2a$10$MeMs8c2Vj/vfTNLj/1WX2uG3dQZxf52WxtY3pIAxE53WxjpQPKdGq | student | 2025-11-27 21:40:45 |
| 13 | aryan | $2a$10$gTxeuI7X1wiUCMLGLXYvdegdjbCZ90ivmC7T6tDTP.cH8hgE4votW | student | 2025-11-27 20:12:18 |
| 14 | eddy24 | $2a$10$jQwzCaXeJJmPGK5hIH6PD.ES/vzLpL6vxKEjN1SI7C85EXrQ3B3X0 | student | 2025-11-23 19:12:47 |
| 15 | anthony24 | $2a$10$fQygaJpIc0Y.eQabRxp8puWUEZXFVWCTsDE6/gqomA7phZ8HEUAh2 | instructor | 2025-11-25 00:24:19 |
| 16 | admin3 | $2a$10$VUfWDCNL.JKYJNiOyam.gupl1ejzWLC/zL70r7aDcfGL2nYwfYI8a | admin | NULL |
+-----+-----+-----+-----+-----+
16 rows in set (0.00 sec)
```

```
MySQL 8.0 Command Line C...
Database changed
mysql> show tables;
+-----+
| Tables_in_erp_db |
+-----+
| courses           |
| enrollments       |
| instructors       |
| scores            |
| sections          |
| settings          |
| students          |
+-----+
7 rows in set (0.00 sec)

mysql> select * from students;
+-----+
| user_id | roll_no | program | year | name |
+-----+
| 5       | 2024000 | CSAI    | 2024 | NULL |
| 8       | 21      | CS      | 2024 | Dennis |
| 10      | 2024041 | CSE     | 2029 | chaitanya |
| 11      | 20240442 | ECE     | 2027 | advik |
| 12      | 2024042 | ec      | 2025 | krishna |
| 13      | 9347239 | cds     | 2026 | aryan |
| 14      | 2024    | CSE     | 2027 | eddy |
+-----+
7 rows in set (0.00 sec)

mysql> select * from instructors;
+-----+
| user_id | name       | department |
+-----+
| 6       | Erling Halaand | CSE |
| 9       | David       | ECE |
| 15      | anthony     | CSE |
+-----+
3 rows in set (0.00 sec)

mysql> |
```

```
MySQL 8.0 Command Line C...
3 rows in set (0.00 sec)

mysql> select * from courses;
+-----+
| code | title | credits | deadline |
+-----+
| CSE210 | IP | 4 | NULL |
| ECE250 | ELD | 4 | NULL |
| MTH201 | LA | 2 | 2025-11-28 |
| MTH203 | W3 | 4 | 2025-11-24 |
| SSH200 | SNS | 4 | 2024-12-23 |
+-----+
5 rows in set (0.00 sec)

mysql> select * from sections;
+-----+
| section_id | course_id | instructor_id | day_time | room | capacity | semester | year |
+-----+
| 4 | ECE250 | 9 | MON 12 | 102 | 498 | 3 | 2026 |
| 5 | MTH201 | 15 | MON 11AM | C101 | 299 | 1 | 2026 |
| 6 | MTH201 | 9 | MON 11AM | 102 | 9 | 1 | 2026 |
| 8 | MTH203 | 15 | Tue 11 | 102 | 230 | 4 | 2026 |
| 13 | CSE210 | 15 | WED | qiugrb | 0 | 4 | 235 |
| 14 | MTH203 | 15 | THU | 201 | 1 | 4 | 2025 |
+-----+
6 rows in set (0.00 sec)

mysql> select * from enrollments;
+-----+
| enrollment_id | student_id | section_id | status |
+-----+
| 1 | 12 | 4 | active |
| 2 | 13 | 4 | active |
| 7 | 13 | 6 | active |
| 10 | 12 | 13 | active |
| 12 | 13 | 14 | active |
| 13 | 12 | 6 | active |
+-----+
6 rows in set (0.00 sec)

mysql> |
```

```
mysql> select * from scores;
+-----+-----+-----+-----+-----+
| section_id | student_id | assessment | score | weight |
+-----+-----+-----+-----+-----+
| 4 | 12 | Midsem | 42 | 40 |
| 4 | 13 | Endsem | 67 | 40 |
| 4 | 13 | Quiz | 78 | 20 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from settings;
+-----+-----+
| key | value |
+-----+-----+
| maintenance_mode | false |
+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

(These databases were used since the starting of the project, therefore some entries used previously do not have all the features, such as last login, deadline etc. The last few entries are working correctly and are amde sure that they align with all the features implemented in our app. Correctly working credentials are mentioned in HOW TO RUN pdf.)

Testing Summary

The University ERP application was tested across all three user roles—Admin, Instructor, and Student—to verify correct behavior, data consistency, and enforcement of system rules. Functional tests confirmed that the Admin can reliably manage users, courses, and sections, including creating and editing records with proper validation such as preventing zero or negative capacities. Maintenance Mode was verified to correctly block Instructor and Student operations while still allowing Admin access. For Instructors, tests validated section viewing, score entry, final grade computation, class statistics retrieval, and exports to CSV, all restricted by maintenance mode and section ownership. Student tests confirmed the full enrollment workflow, including browsing the course catalog, registering and dropping courses, enforcing capacity limits, preventing duplicate or cross-section enrollment, and blocking registration after course deadlines. Timetable and grade viewing behaved consistently, and CSV export produced correct files.

Across all flows, UI elements—tables, forms, animated maintenance banners, and stats—responded dynamically to data changes. Errors such as invalid input, missing IDs, or deadline violations were handled gracefully with user-facing messages. The system maintained referential integrity between the two databases and performed updates (such as capacity increases and decreases) reliably. Overall, end-to-end testing demonstrated that the system meets expected functionality, enforces rules strictly, handles edge cases properly, and provides a stable and coherent experience across all roles.